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Kali Linux (Operator Workstation)

Objective

This document describes the role, configuration, and validation of the Kali Linux virtual machine that serves as the operator workstation within the Red Team lab.

The objective is to establish a dedicated workstation from which offensive operations are conducted while ensuring that the operator's infrastructure remains logically separated from the target environment. **This separation reflects the principle that an operator should minimise direct exposure of their working environment during an engagement.**

Role Within the Lab

The Kali Linux virtual machine functions as the operator's workstation throughout all phases of the engagement.

Its primary responsibilities include:

- managing offensive tooling;
- hosting and managing the Command and Control (C2) framework;
- generating payloads for simulated initial access;
- establishing secure management connections to the Ubuntu Redirector;
- issuing commands to compromised systems through the redirector;
- collecting and analysing information obtained during post-exploitation activities.

The operator workstation is not intended to communicate directly with the Windows target. All operational traffic destined for the target environment must be relayed through the Ubuntu Redirector.

Design Considerations

The operator workstation is intentionally isolated from the target network to simulate the separation commonly maintained between an operator's infrastructure and the engagement environment.

This design provides several advantages:

- reduces direct exposure of the operator's infrastructure;
- encourages the use of intermediary infrastructure during operations;
- reinforces the practice of network segmentation;
- provides a controlled environment for learning tunnelling and traffic redirection techniques.

Although this lab is conducted within a virtualised environment, the architecture is designed to encourage operational habits that can be applied to larger and more realistic infrastructures.

Configuration

The Kali Linux virtual machine had already been deployed before the creation of this documentation. Consequently, this document focuses on validating and documenting its operational role within the lab rather than detailing the operating system installation process.

The workstation was reviewed to ensure it aligns with the objectives of the lab and can perform the tasks required during subsequent phases of the engagement.

Validation

The following validation checks should be completed before progressing to the next stage of the lab:

- verify that the operator workstation is functioning correctly;
- verify that the workstation can establish an SSH session with the Ubuntu Redirector;
- verify that the workstation cannot communicate directly with the Windows target;
- verify that the workstation is capable of hosting the selected C2 framework;
- create or update a clean snapshot after successful validation.

Any issues identified during validation should be resolved before additional lab components are configured.

Observations

This section should record notable findings encountered during configuration or validation.

Examples include configuration changes, unexpected behaviour, networking observations, or decisions made to improve the overall lab design.

Questions

This section captures questions that arise during the setup process together with their corresponding answers or conclusions.

Lessons Learned

Summarise the key concepts gained from configuring the operator workstation.

This section should focus not only on what was configured but also on why the chosen configuration supports the objectives of the lab and how it contributes to future phases of the engagement.

Changes Made

Date	Change	Reason
29/07/2026	Initial documentation created	Establish baseline configuration